



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-1.2

Student Name: Vaibhav Kumar

UID: 23BCS13386

Branch: B.E-C.S.E

Section/Group: 23KRG-2B

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Subject Name: DAA

Subject Code: 23CSH-301

1. **Aim:** Code implement power function in $O(\log n)$ time complexity.
2. **Objective:** To implement power function in $O(\log n)$ time complexity.
3. **Input/Apparatus Used:** In this program , power is divided by 2 in order to get complexity in log.
4. **Procedure/Algorithm: Pseudocode:**

```
if(y==0):  
    return 1;  
temp = power(x,y/2); if(y%2==0):  
    return temp*temp; else  
    return x*temp*temp;
```

Various possible cases

- 1)The power value can be 0
- 2)The power value can be negative.
- 3)The power value can be even.
- 4)The power value can be odd.

5. Code:

```
#include <iostream>
using namespace std;

double power(double x, int y) {
    if (y == 0)
        return 1;

    double temp = power(x, y / 2);

    if (y % 2 == 0)
        return temp * temp;
    else {
        if (y > 0)
            return x * temp * temp;
        else
            return (temp * temp) / x;
    }
}

int main() {
    double x;
    int y;

    cout << "Enter base (x): ";
    cin >> x;
    cout << "Enter exponent (y): ";
    cin >> y;

    cout << x << "^" << y << " = " << power(x, y) << endl;

    return 0;
}
```

6. Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\C tutorial 2> ./DAAExp2.exe
Enter base (x): 10
Enter exponent (y): 5
10^5 = 100000
PS C:\Users\vaibh\OneDrive\Desktop\C tutorial 2> 
```